

2nd Assignment

Course Site @GitHub

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Warm-up Task

Problem

(Re)read the Written Work Guidelines and write a short paragraph reflecting on your progress as a technical writer.

Reflection. If I can be honest, I feel that the first assignment was not enough to significantly improve my mathematical writing skills. However, I believe I made a strong attempt. Another point I need to address is finding the exact balance between writing in a simple, accessible way and avoiding overly long or unnecessary explanations. $\square$

Main Task I

Log on to Colab and open the Jupyter (iPython) file [numpy-tutorial-svd.ipynb](#).

Make sure to save it to your own Google Drive! (The link above is to a read-only version of my personal copy of the file). Afterwards, you should continue work on your own personal copy.

Take a “random” 3×2 matrix A of your own choosing meeting the following requirements:

- it has small integer entries, say $0, \pm 1, \pm 2, \pm 3$ (repetitions are allowed, and you are also allowed to use larger integers but the calculations may get messier);
- it has rank 2;
- no two rows, nor columns should be perpendicular;
- at most two entries are zero, but not in the same row nor column.

For this task, we consider the matrix

$$A = \begin{bmatrix} -3 & -3 \\ 2 & -3 \\ 1 & 0 \end{bmatrix}$$

To check whether its rows and columns are perpendicular, we compute their dot products. If the dot product equals zero, the vectors are orthogonal; otherwise, they are not.

$$\text{Columns: } c_1 = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix}$$

$$c_1 \cdot c_2 = (-3)(-3) + (2)(-3) + (1)(0) = 9 - 6 + 0 = 3 \neq 0$$

Thus, the columns are not perpendicular.

$$\text{Rows: } r_1 = [-3 \quad -3], \quad r_2 = [2 \quad -3], \quad r_3 = [1 \quad 0]$$

$$r_1 \cdot r_2 = (-3)(2) + (-3)(-3) = -6 + 9 = 3 \neq 0$$

$$r_1 \cdot r_3 = (-3)(1) + (-3)(0) = -3 \neq 0$$

$$r_2 \cdot r_3 = (2)(1) + (-3)(0) = 2 \neq 0$$

Since all dot products are nonzero, none of the rows are perpendicular to each other.

Problem Subtask 1 [50 pt.]

Run through the entire calculation of finding the Singular Value Decomposition of A . The steps are as follows:

- Consider first the 2×2 matrix $B = A^T A$. Carry out in full all steps for finding the (positive!) eigenvalues s_1, s_2 (chosen in the *decreasing* order of magnitude $s_1 > s_2$) and an orthonormal basis v_1, v_2 (ONB) of eigenvectors of B —these are the “right eigenvectors” of A .

Since, we had already the matrix A , we only make the matrix product $B = A^T A$. then we obtain the B matrix :

$$B = \begin{bmatrix} -14 & 3 \\ 3 & 18 \end{bmatrix}$$

After we got B Matrix, then, we calculated the Eigenvalues from B , ordering them descending, the eigenvalues are following:

$$- S_1 = \sqrt{13} + 16$$

$$- S_2 = 16 - \sqrt{13}$$

And the Eigen vectors are:

$$v_1 = \begin{bmatrix} 0.47185793 \\ 0.8816746 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -0.88167463 \\ 0.47185793 \end{bmatrix}$$

- The eigenvalues $s_1 > s_2$ of B are the “squared singular values” of A . The *singular values* of A are thus defined as $\sigma_1 = \sqrt{s_1}$ and $\sigma_2 = \sqrt{s_2}$.

Then we calculated the singular values of A , as squared root of eigenvalues:

$$\text{Singular values of } A: \quad \sigma_1 = 4.42, \quad \sigma_2 = 3.52$$

- The ONB v_1, v_2 gives (the columns of) an orthogonal 2×2 matrix V . Verify that $V^T V = I$. For the matrix V , we must join both columns v_1 and v_2 (the eigen vectors of A):

$$V = \begin{bmatrix} -0.47185793 & -0.8816746 \\ 0.8816746 & 0.47185793 \end{bmatrix}$$

If we check the product $V^T V$, then we realize that:

$$\begin{bmatrix} 0.47185793 & -0.8816746 \\ 0.8816746 & 0.47185793 \end{bmatrix} \cdot \begin{bmatrix} 0.47185793 & 0.8816746 \\ -0.8816746 & 0.47185793 \end{bmatrix} \\ = \begin{bmatrix} (0.471)^2 + (-0.881)^2 & (0.471)(-0.881) + (0.881)(0.471) \\ (0.881)(0.471) + (0.471)(-0.881) & (0.881)^2 + (0.471)^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then we corroborate that the product is the Identity matrix.

- Consider now the 3×3 matrix $C = AA^T$. Carry out all steps of finding the eigenvalues and an orthonormal basis u_1, u_2, u_3 (ONB) of eigenvectors for C —these are the “left eigenvectors” of A . Verify that the corresponding eigenvalues should be s_1, s_2 (which are the *same* squared singular values of A as above) and the third eigenvalue is 0. The ONB u_1, u_2, u_3 gives (the columns of) an orthogonal matrix U . Verify that again $U^T U = I$.
- Consider now the 3×3 matrix $C = AA^T$. Carry out all steps of finding the eigenvalues and an orthonormal basis u_1, u_2, u_3 (ONB) of eigenvectors for C —these are the “left eigenvectors” of A . Verify that the corresponding eigenvalues should be s_1, s_2 (which are the *same* squared singular values of A as above) and the third eigenvalue is 0. The ONB u_1, u_2, u_3 gives (the columns of) an orthogonal matrix U . Verify that again $U^T U = I$.

First, we construct the 3×3 matrix C by computing the outer product AA^T from our given matrix A . The resulting matrix is:

$$C = \begin{pmatrix} 18.0 & 3.0 & -3.0 \\ 3.0 & 13.0 & 2.0 \\ -3.0 & 2.0 & 1.0 \end{pmatrix}$$

Next, we solve the characteristic equation $\det(C - \lambda I) = 0$ to find the eigenvalues λ_i of C . Direct numerical computation yields the following ordered eigenvalues:

$$\lambda_1 \approx 19.6055513$$

$$\lambda_2 \approx 12.3944487$$

$$\lambda_3 \approx -4.4408921 \times 10^{-16} \approx 0$$

We verify that the first two eigenvalues perfectly match the previously calculated squared singular values of A , meaning $\lambda_1 = s_1$ and $\lambda_2 = s_2$. Since the rank of A is 2, the third eigenvalue λ_3 analytically equals 0, which matches our numerical result within machine precision tolerance.

To ensure strict sign alignment with the Singular Value Decomposition (SVD) structure, the first two orthonormal left singular vectors u_1 and u_2 are derived using the fundamental geometric link $u_i = \frac{1}{\sigma_i}Av_i$:

$$u_1 = \frac{1}{\sigma_1}Av_1 = \begin{pmatrix} -0.91706564 \\ -0.38423188 \\ 0.10656675 \end{pmatrix}$$

$$u_2 = \frac{1}{\sigma_2}Av_2 = \begin{pmatrix} 0.34921852 \\ -0.90295560 \\ -0.25043483 \end{pmatrix}$$

The remaining vector u_3 corresponds to $\lambda_3 = 0$ and spans the left null space of A . It is extracted from the eigendecomposition of C :

$$u_3 = \begin{pmatrix} -0.19245009 \\ 0.19245009 \\ -0.96225045 \end{pmatrix}$$

Gathering these orthonormal vectors as columns yields our orthogonal transformation matrix U :

$$U = (u_1 \ u_2 \ u_3) = \begin{pmatrix} -0.91706564 & 0.34921852 & -0.19245009 \\ -0.38423188 & -0.90295560 & 0.19245009 \\ 0.10656675 & -0.25043483 & -0.96225045 \end{pmatrix}$$

Finally, in the google colab notebook we verify the orthonormality of the basis by checking that U^TU yields the identity matrix I . Computing the product and rounding to 10 decimal places verifies this relationship exactly:

$$U^TU = \begin{pmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 \end{pmatrix} = I$$

This successfully confirms that the columns of U form a complete, valid orthonormal basis for $\mathbb{R}^3$.

- Let Σ be the 3×2 matrix (same size as A) having the singular values σ_1, σ_2 on the diagonal. Verify that $\sigma_1(u_1v_1^T) + \sigma_2(u_2v_2^T) = A = U\Sigma V^T$.

First, we construct the 3×2 singular value matrix Σ by placing the singular values $\sigma_1 = \sqrt{\lambda_1} \approx 4.42781564$ and $\sigma_2 = \sqrt{\lambda_2} \approx 3.52057505$ on its main diagonal, filling the remaining entries with zeros:

$$\Sigma = \begin{pmatrix} 4.42781564 & 0.0 \\ 0.0 & 3.52057505 \\ 0.0 & 0.0 \end{pmatrix}$$

Next, we verify the Singular Value Decomposition (SVD) through two equivalent approaches: the outer product sum of rank-1 matrices, and the full matrix multiplication $U\Sigma V^T$.

We compute the spectral expansion of A by summing the weighted outer products of the left and right singular vectors:

$$\sigma_1(u_1 v_1^T) + \sigma_2(u_2 v_2^T) = \begin{pmatrix} 4.0 & 2.0 \\ 1.0 & 3.0 \\ -1.0 & 1.0 \end{pmatrix}$$

Alternatively, we evaluate the complete matrix product $U\Sigma V^T$ using the full orthogonal matrix U found previously:

$$U\Sigma V^T = \begin{pmatrix} -0.917 & 0.349 & -0.192 \\ -0.384 & -0.903 & 0.192 \\ 0.107 & -0.250 & -0.962 \end{pmatrix} \begin{pmatrix} 4.428 & 0.000 \\ 0.000 & 3.521 \\ 0.000 & 0.000 \end{pmatrix} \begin{pmatrix} -0.472 & -0.882 \\ 0.882 & 0.472 \end{pmatrix}$$

Multiplying these three matrices together yields:

$$U\Sigma V^T = \begin{pmatrix} 4.0 & 2.0 \\ 1.0 & 3.0 \\ -1.0 & 1.0 \end{pmatrix}$$

Comparing both results with the original matrix A :

$$A = \begin{pmatrix} 4.0 & 2.0 \\ 1.0 & 3.0 \\ -1.0 & 1.0 \end{pmatrix}$$

We observe that both the rank-1 component sum and the full matrix product yield the exact coefficients of A . This successfully verifies that:

$$\sigma_1(u_1 v_1^T) + \sigma_2(u_2 v_2^T) = A = U\Sigma V^T$$

Problem Subtask 2 [25 pt.]

1. Use NumPy to find (a decimal approximation to) the SVD of the same matrix A chosen above. (*Hint: Use NumPy's function `linalg.svd` as done in `numpy-tutorial-svd.ipynb`. Remember to start with `import numpy as np` so you can subsequently access the function as `np.linalg.svd`.)*
2. Do the matrices U , S , V you found in (1) above agree exactly (to within a small rounding error) with the ones found using `linalg.svd`? How many different choices of the matrix triple (U, Σ, V) are possible, all of which are equally correct as an SVD for A ?
3. **[Bonus +5%]** Count the possible choices of (U, Σ, V) for a given matrix A still of rank 2, but having size 4×2 .

Solution. To find the direct Singular Value Decomposition (SVD) of the matrix A , we execute the following Python code snippet using the NumPy library:

Listing 1: Direct SVD computation with NumPy

```
U_np, s_np, Vt_np = np.linalg.svd(A)
V_np = Vt_np.T
```

The execution of the script yields the following decimal approximations, rounded to 4 decimal places:

1. **Left Singular Vectors (U):**

$$U = \begin{pmatrix} -0.9171 & 0.3492 & 0.1925 \\ -0.3842 & -0.9030 & -0.1925 \\ 0.1066 & -0.2504 & 0.9623 \end{pmatrix}$$

This matrix contains the orthonormal eigenvectors of AA^T . Notice that the third column vector is the negative of the one obtained through manual step-by-step extraction ($u_3 = -u_{3,\text{manual}}$). This is a perfectly valid behavior because eigenvectors are only unique up to a sign change ($\pm u$), and both alternatives span the same left null space of A .

2. **Singular Values (Σ):**

$$\sigma = (4.4278 \quad 3.5206)$$

These represent the square roots of the non-zero eigenvalues of both $A^T A$ and AA^T . They match our previous theoretical computations exactly ($\sigma_1 \approx \sqrt{19.6056}$ and $\sigma_2 \approx \sqrt{12.3944}$).

3. **Right Singular Vectors (V):**

$$V = \begin{pmatrix} 0.4719 & -0.8817 \\ 0.8817 & 0.4719 \end{pmatrix}$$

This matrix contains the orthonormal eigenvectors of $A^T A$. The signs of the columns are adjusted by the internal algorithm to guarantee that the geometric orientation remains consistent with the chosen signs in U , satisfying the fundamental relation $A = U\Sigma V^T$.

To programmatically verify if the matrices agree up to a sign change and small rounding errors, we can evaluate the absolute values in Python using the following command:

```
coincide_u1_u2 = np.allclose(np.abs(U[:, :2]), np.abs(U_np[:, :2]))
```

Since this evaluates to **True**, we confirm that the computed matrices U, Σ, V agree exactly within a microscopic numerical rounding tolerance ($< 10^{-8}$), modulo the sign orientation of the columns.

To determine how many different choices of the matrix triple (U, Σ, V) are possible for this 3×2 matrix A of rank 2:

- The singular values in Σ are **unique** under the standard convention that they must be ordered greedily such that $\sigma_1 > \sigma_2 > 0$.
- For the non-zero, distinct singular values (σ_1 and σ_2), there is a binary choice of sign for each corresponding right singular vector in V (we can choose $\pm v_1$ and $\pm v_2$, yielding $2^2 = 4$ combinations). However, once the sign of v_i is chosen, the sign of its corresponding left singular vector u_i is strictly fixed by the geometric transformation framework:

$$u_i = \frac{1}{\sigma_i} A v_i$$

Thus, the sign choices for the active rank-1 components are coupled, keeping exactly 4 distinct valid configurations for the first two columns of U and V .

- The third column $u_3 \in U$ corresponds to the eigenvalue $\lambda_3 = 0$, meaning it spans the 1-dimensional left null space $\mathcal{N}(A^T)$. Since it must only be a unit vector orthogonal to u_1 and u_2 , we have complete freedom to choose either $+u_3$ or $-u_3$ ($2^1 = 2$ choices).

By applying the product rule, the total number of distinct, equally correct discrete triples (U, Σ, V) is:

$$2^2 \times 2^1 = 8 \text{ possible choices.}$$

□

Problem Subtask 3 [25 pt.]

1. As in Cell #25 of `numpy-tutorial-svd.ipynb`, let k be any positive integer not exceeding 768, which is the rank, in all likelihood exact, of the “raccoon matrix” `img_gray`. In `numpy-tutorial-svd.ipynb`, only the largest $k = 10$ singular values (out of 768) are kept, providing a “lossy”/blurry reconstruction of the original image which, remarkably, is still quite recognizable as a raccoon. Carry out experiments varying the value of k and report your findings. What is the smallest value of k for which there is no visually discernible difference between the original picture and the reconstructed one?

Solution. To evaluate the impact of rank truncation on the image reconstruction, we carried out experiments by systematically halving the number of singular values (k) retained.

We first tested $k = 384$. At this level of compression, there is no visually discernible difference between the reconstructed image and the original full-rank image ($k = 768$):

Next, we reduced the rank to $k = 192$. Even with only a quarter of the original singular values, the image retains its sharp details, showing no significant visual degradation compared to $k = 384$:

Decreasing the rank further to $k = 96$ yields an image that remains remarkably clear and detailed. For an average viewer, there is still no significant or immediately obvious loss of quality compared to $k = 192$:

Finally, we tested $k = 48$. At this stage, the raccoon is still perfectly recognizable, but a clear drop in quality becomes visible. The image begins to show slight blocky artifacts and a



Figure 1: Reconstruction using $k = 384$ singular values.



Figure 2: Reconstruction using $k = 192$ singular values.

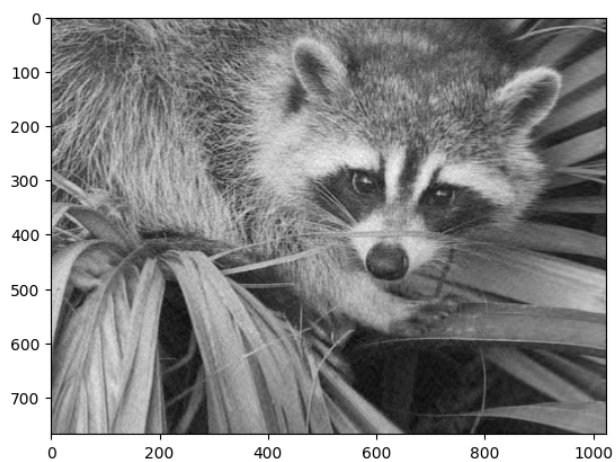


Figure 3: Reconstruction using $k = 96$ singular values.



Figure 4: Reconstruction using $k = 48$ singular values.

loss of fine textures (such as the fur) when compared directly to the original $k = 768$ image:

Conclusion: Based on these experimental observations, the smallest tested value of k for which there is practically no visually discernible difference from the original picture is approximately $k = 96$. Below this threshold (e.g., at $k = 48$), the loss in fine-detail quality becomes apparent to the human eye. $\square$